

JT_DEV_B51_1.7

PHP blog

1 Initial situation

- 1.1 You have now acquired the basics of the PHP language. To practice them, you will now create a blog web application with a simple login system and a list of blog posts. Unlike previous TypeScript and React todo list apps, which were « frontend only » apps (all the code ran in the browser, you just used a REST API to fetch data). This time, you will create a « back-end only » app, where all the code runs on the server with PHP and only static HTML and CSS are sent to the browser. No JavaScript is used in this app.
- 1.2 This kind of app architecture was the most common one before the rise of front-end frameworks like React, Vue or Angular. You may find it old-fashioned, but it's still widely used today for small to medium web applications, as it's simple to implement and maintain. This kind of architecture is also really similar to API backends that are really common today, so it's important to understand how it works.
- 1.3 With this app, you will learn how to upload images with an HTML form and PHP, how to verify user passwords securely and how to check if the user is logged in with PHP session. You'll also learn to use a relational database to store the information of your app and how to connect to it from PHP with PDO.

2 Skills from the ICT training plan worked on

Skills ranked by importance of implementation in the project: [consult the training plan](#).

- 2.1 g5 : Implémenter les applications et les interfaces selon le concept en respectant les exigences de sécurité.
- 2.2 c2 : Mettre en œuvre des modèles de données dans un dispositif de stockage de données numériques.
- 2.3 g3 : Evaluer et documenter la sécurité des applications et des interfaces.
- 2.4 c3 : Planifier, implémenter et documenter la sécurité et la protection des données pour des solutions ICT.
- 2.5 a4 : Planifier les projets ICT et les tâches selon un modèle de procédure.
- 2.6 a6 : Vérifier l'avancement des projets ICT et des tâches et en faire état selon le modèle de procédure.
- 2.7 g6 : Vérifier la qualité et la sécurité des applications et des interfaces.
- 2.8 g2 : Vérifier la faisabilité technique des prototypes d'interfaces utilisateur et les développer.
- 2.9 a7 : Remettre la solution ICT au client et clôturer le projet.

3 Deliverables

- 3.1 GitHub repository containing the working app.
- 3.2 GitHub project with all tasks completed.

4 Andragogic conditions

- 4.1 Work is carried out independently, on an individual basis.
- 4.2 You are allowed to help each other, share ideas and discuss solutions, but each trainee must provide their own implementation with their own code.
- 4.3 Apprentices have access to all the resources they need, particularly on the web.



- 4.4 Don't hesitate to ask AI for explanations, but avoid asking it for answers. The more you ask for answers, the less you learn.
- 4.5 **Maximum allowed time for the project: 10 full days (7.5 hours per day).** You must manage your time independently. On project beginning, inform your trainer of the start date and time.

5 Steps

- 5.1 Create a new GitHub Project from [the PHP blog template](#) to your account (Use the « ... » then « Make a copy » menu). Select your account to host the Project on and name the project with your GitHub pseudo, for example `php-blog-<your-username>`. **Do not forget to tick the « Draft issues will be copied if selected » checkbox!**
- 5.2 Ensure you have [PHP](#) and [Composer](#) installed on your computer (install recent versions like PHP 8.5+ and Composer 2+).
- 5.3 Setup your project by following instructions in the « Project setup » task.
- 5.4 Implement all the tasks described in the project.

6 Resources

- 6.1 Simple PHP blog example <https://blog.in.jt-lab.ch/> (spoiler: there is bugs and security issues in this app, would you find them ?)
- 6.2 <https://grafikart.fr/tutoriels/php>
- 6.3 <https://www.php.net/docs.php>
- 6.4 <https://phpdelusions.net/>
- 6.5 <https://laravel.com/docs/12.x/pint>
- 6.6 PHP session : <https://grafikart.fr/tutoriels/session-php-1128>
- 6.7 Access to SQL database with PHP PDO : <https://grafikart.fr/tutoriels/pdo-php-1141>
- 6.8 Manage html forms with php : <https://grafikart.fr/tutoriels/forms-php-1123>
- 6.9 Simple login system with PHP : <https://grafikart.fr/tutoriels/tp-login-vues-1131>

7 Evaluation criteria

- 7.1 All the tasks from the GitHub project have been implemented.
- 7.2 For each task there is a pull request, with a reviewer and a [close keyword](#).
- 7.3 The review comments have been taken into account in the pull request.
- 7.4 All the commits must comply with the [conventional commits](#) standard.
- 7.5 The codebase must be formatted with pint (psr12 style).
- 7.6 The app must work out of the box `php -S localhost:8080 -t ./public`.
- 7.7 Your codebase respects the security recommendation (protected against sql injections, XSS, and right password hash storage and verification)
- 7.8 All errors are managed on all pages and conveniently displayed to the user.
- 7.9 The README contains a brief description of the app, setup instructions and a list of features.
- 7.10 The style of the app is clean and modern. Use recent CSS features.
- 7.11 All features work as described in the project tasks.
- 7.12 There is no dead code.

